

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad 2(a + 3b) =$$

$$(2) \quad 5(4x + 7) =$$

$$(3) \quad 2(4x - 3) =$$

$$(4) \quad 4(2y + 1) =$$

$$(5) \quad 4(3a + 1x) =$$

$$(6) \quad 5(a + 2y) =$$

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad 4(b - 2) =$$

$$(2) \quad -2(x + 1a) =$$

$$(3) \quad 3(2y + 1) =$$

$$(4) \quad 3(a + 6) =$$

$$(5) \quad 5(2x + 3y) =$$

$$(6) \quad 4(y - 4) =$$

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad 2(b + 2x) =$$

$$(2) \quad -2(x + 1y) =$$

$$(3) \quad -4(3a - 4b) =$$

$$(4) \quad -5(x - 1a) =$$

$$(5) \quad 5(a - 6) =$$

$$(6) \quad 5(y + 4) =$$

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad 5(2x + 2) =$$

$$(2) \quad 2(3y - 4b) =$$

$$(3) \quad 4(2a + 2) =$$

$$(4) \quad 2(3a - 1) =$$

$$(5) \quad 5(2x + 5) =$$

$$(6) \quad 2(x - 1b) =$$

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad 4(x - 3a) =$$

$$(2) \quad 2(3b + 8) =$$

$$(3) \quad 4(4b + 4) =$$

$$(4) \quad 4(2a + 4) =$$

$$(5) \quad 3(3a + 2) =$$

$$(6) \quad 4(y - 3a) =$$

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad 4x + 4(4x + 1y) =$$

$$(2) \quad 4(x + 2) + 3(x - 5) =$$

$$(3) \quad 4(y + 3) + 3(y - 5) =$$

$$(4) \quad 2(y + 4) + 4(y + 1) =$$

$$(5) \quad 4(a + 4) + 4(a + 4) =$$

$$(6) \quad 3(x + 1) + 4(x - 1) =$$

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad 4(b + 2) + 4(b + 2) =$$

$$(2) \quad 3a + 4(4a + 1y) =$$

$$(3) \quad 4y + 3(-1y + 2x) =$$

$$(4) \quad 4(x + 4) + 4(x - 4) =$$

$$(5) \quad 4(x + 4) + 2(x - 2) =$$

$$(6) \quad 2a + 2(3a + 3y) =$$

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad 4a + 4(-1a + 4b) =$$

$$(2) \quad 2y + 3(-1y + 2b) =$$

$$(3) \quad 2y + 3(-3y + 2x) =$$

$$(4) \quad 3(b + 1) + 4(b + 3) =$$

$$(5) \quad 4(a + 1) + 2(a - 1) =$$

$$(6) \quad 4(a + 2) + 3(a + 5) =$$

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad 3(y + 2) + 2(y + 3) =$$

$$(2) \quad 5(a + 4) + 4(a + 3) =$$

$$(3) \quad 4(a + 4) + 3(a + 1) =$$

$$(4) \quad 2(x + 3) + 2(x + 2) =$$

$$(5) \quad 4y + 3(-1y + 1a) =$$

$$(6) \quad 3(b + 4) + 2(b - 5) =$$

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (a + 3)(a - 3) =$$

$$(2) \quad (y + 4)^2 =$$

$$(3) \quad (x - 4)^2 =$$

$$(4) \quad (x - 3)^2 =$$

$$(5) \quad (a + 4)(a - 4) =$$

$$(6) \quad (a - 7)^2 =$$

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (a + 3)(a - 3) =$$

$$(2) \quad (a + 6)^2 =$$

$$(3) \quad (x + 8)^2 =$$

$$(4) \quad (a + 7)^2 =$$

$$(5) \quad (a - 6)^2 =$$

$$(6) \quad (y - 2)^2 =$$

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad 5a + 3(2a + 1b) =$$

$$(2) \quad 3y + 3(3y + 1a) =$$

$$(3) \quad 5(x + 1) + 2(x - 4) =$$

$$(4) \quad 4y + 3(3y + 4x) =$$

$$(5) \quad 5(x + 1) + 4(x + 1) =$$

$$(6) \quad 3(a + 4) + 4(a + 5) =$$

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (y + 6)^2 =$$

$$(2) \quad (a + 4)^2 =$$

$$(3) \quad (y + 7)(y - 7) =$$

$$(4) \quad (y - 2)^2 =$$

$$(5) \quad (x - 4)^2 =$$

$$(6) \quad (a - 6)^2 =$$

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (a - 6)^2 =$$

$$(2) \quad (x + 3)^2 =$$

$$(3) \quad (x + 6)^2 =$$

$$(4) \quad (y + 8)^2 =$$

$$(5) \quad (x + 8)(x - 8) =$$

$$(6) \quad (a - 5)^2 =$$

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (y + 6)(y + 6) =$$

$$(2) \quad 4b(3b - 4) =$$

$$(3) \quad (x - 4)(x + 2) =$$

$$(4) \quad 4b(1b + 1) =$$

$$(5) \quad 3a(2a + 6) =$$

$$(6) \quad 3y(1y - 3x) =$$

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (y + 3)(y - 3) =$$

$$(2) \quad (x + 5)^2 =$$

$$(3) \quad (y - 2)^2 =$$

$$(4) \quad (y + 4)(y - 4) =$$

$$(5) \quad (x + 3)(x - 3) =$$

$$(6) \quad (x + 2)(x - 2) =$$

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad 3b(3b + 6x) =$$

$$(2) \quad 5b(3b - 2y) =$$

$$(3) \quad 4x(1x - 3) =$$

$$(4) \quad 2x(3x + 3b) =$$

$$(5) \quad (x + 3)(x - 5) =$$

$$(6) \quad 4x(2x - 3a) =$$

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad 4y(3y + 6) =$$

$$(2) \quad 4b(2b + 3y) =$$

$$(3) \quad (y - 3)(y + 1) =$$

$$(4) \quad 4b(3b - 4y) =$$

$$(5) \quad (y - 6)(y - 5) =$$

$$(6) \quad 2y(3y + 4x) =$$

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (y - 1)(y - 2) =$$

$$(2) \quad (a + 2)(a + 5) =$$

$$(3) \quad 3a(1a - 1) =$$

$$(4) \quad 3a(2a + 6) =$$

$$(5) \quad 5x(3x - 3) =$$

$$(6) \quad (a - 3)(a + 5) =$$

Section 2. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad 3y(4y - 4a) =$$

$$(2) \quad 3x(2x + 4) =$$

$$(3) \quad (x + 4)(x + 6) =$$

$$(4) \quad 2y(1y + 3) =$$

$$(5) \quad (a - 2)(a + 5) =$$

$$(6) \quad (y + 6)(y + 3) =$$

Section 2 - Solutions

11a-1: $a^2 - 16$

11a-2: $x^2 + 12x + 36$

11a-3: $x^2 - 36$

11a-4: $a^2 - 9$

11a-5: $x^2 + 6x + 9$

11a-6: $x^2 + 14x + 49$

11b-1: $a^2 - 16$

11b-2: $x^2 + 12x + 36$

11b-3: $x^2 - 36$

11b-4: $a^2 - 9$

11b-5: $x^2 + 6x + 9$

11b-6: $x^2 + 14x + 49$

12a-1: $a^2 - 16$

12a-2: $x^2 + 12x + 36$

12a-3: $x^2 - 36$

Section 2 - Solutions (continued)

13b-1: $a^2 - 16$

13b-2: $x^2 + 12x + 36$

13b-3: $x^2 - 36$

13b-4: $a^2 - 9$

13b-5: $x^2 + 6x + 9$

13b-6: $x^2 + 14x + 49$

14a-1: $a^2 - 16$

14a-2: $x^2 + 12x + 36$

14a-3: $x^2 - 36$

14a-4: $a^2 - 9$

14a-5: $x^2 + 6x + 9$

14a-6: $x^2 + 14x + 49$

14b-1: $a^2 - 16$

14b-2: $x^2 + 12x + 36$

14b-3: $x^2 - 36$

Section 2 - Solutions (continued)

- 14b-4: $a^2 - 9$
- 14b-5: $x^2 + 6x + 9$
- 14b-6: $x^2 + 14x + 49$
- 15a-1: $a^2 - 16$
- 15a-2: $x^2 + 12x + 36$
- 15a-3: $x^2 - 36$
- 15a-4: $a^2 - 9$
- 15a-5: $x^2 + 6x + 9$
- 15a-6: $x^2 + 14x + 49$
- 15b-1: $a^2 - 16$
- 15b-2: $x^2 + 12x + 36$
- 15b-3: $x^2 - 36$
- 15b-4: $a^2 - 9$
- 15b-5: $x^2 + 6x + 9$
- 15b-6: $x^2 + 14x + 49$

Section 2 - Solutions (continued)

- 12a-4: $a^2 - 9$
- 12a-5: $x^2 + 6x + 9$
- 12a-6: $x^2 + 14x + 49$
- 12b-1: $a^2 - 16$
- 12b-2: $x^2 + 12x + 36$
- 12b-3: $x^2 - 36$
- 12b-4: $a^2 - 9$
- 12b-5: $x^2 + 6x + 9$
- 12b-6: $x^2 + 14x + 49$
- 13a-1: $a^2 - 16$
- 13a-2: $x^2 + 12x + 36$
- 13a-3: $x^2 - 36$
- 13a-4: $a^2 - 9$
- 13a-5: $x^2 + 6x + 9$
- 13a-6: $x^2 + 14x + 49$

Section 2 - Solutions (continued)

16a-1: $a^2 - 16$

16a-2: $x^2 + 12x + 36$

16a-3: $x^2 - 36$

16a-4: $a^2 - 9$

16a-5: $x^2 + 6x + 9$

16a-6: $x^2 + 14x + 49$

16b-1: $a^2 - 16$

16b-2: $x^2 + 12x + 36$

16b-3: $x^2 - 36$

16b-4: $a^2 - 9$

16b-5: $x^2 + 6x + 9$

16b-6: $x^2 + 14x + 49$

17a-1: $a^2 - 16$

17a-2: $x^2 + 12x + 36$

17a-3: $x^2 - 36$

Section 2 - Solutions (continued)

18b-1: $a^2 - 16$

18b-2: $x^2 + 12x + 36$

18b-3: $x^2 - 36$

18b-4: $a^2 - 9$

18b-5: $x^2 + 6x + 9$

18b-6: $x^2 + 14x + 49$

19a-1: $a^2 - 16$

19a-2: $x^2 + 12x + 36$

19a-3: $x^2 - 36$

19a-4: $a^2 - 9$

19a-5: $x^2 + 6x + 9$

19a-6: $x^2 + 14x + 49$

19b-1: $a^2 - 16$

19b-2: $x^2 + 12x + 36$

19b-3: $x^2 - 36$

Section 2 - Solutions (continued)

- 19b-4: $a^2 - 9$
- 19b-5: $x^2 + 6x + 9$
- 19b-6: $x^2 + 14x + 49$
- 20a-1: $a^2 - 16$
- 20a-2: $x^2 + 12x + 36$
- 20a-3: $x^2 - 36$
- 20a-4: $a^2 - 9$
- 20a-5: $x^2 + 6x + 9$
- 20a-6: $x^2 + 14x + 49$
- 20b-1: $a^2 - 16$
- 20b-2: $x^2 + 12x + 36$
- 20b-3: $x^2 - 36$
- 20b-4: $a^2 - 9$
- 20b-5: $x^2 + 6x + 9$
- 20b-6: $x^2 + 14x + 49$

Section 2 - Solutions (continued)

- 17a-4: $a^2 - 9$
- 17a-5: $x^2 + 6x + 9$
- 17a-6: $x^2 + 14x + 49$
- 17b-1: $a^2 - 16$
- 17b-2: $x^2 + 12x + 36$
- 17b-3: $x^2 - 36$
- 17b-4: $a^2 - 9$
- 17b-5: $x^2 + 6x + 9$
- 17b-6: $x^2 + 14x + 49$
- 18a-1: $a^2 - 16$
- 18a-2: $x^2 + 12x + 36$
- 18a-3: $x^2 - 36$
- 18a-4: $a^2 - 9$
- 18a-5: $x^2 + 6x + 9$
- 18a-6: $x^2 + 14x + 49$